

Information Theory & Error Correcting Codes

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Disclaimer

This document contains the lectures given by Dr. Seffah.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 0

Remainders

1. Congruences & $\mathbb{Z}/m\mathbb{Z}$ Arithmetic

Definition 1.1 (Congruence Of Integers / Congruence Class): Let $a, b, m \in \mathbb{Z}$ with $m > 0$, we say that a is congruent to b modulo m and we write $a \equiv b \pmod{m}$ if $m \mid a - b$ which gives an equivalence relation. The class of a in the congruence relation by m is called the congruence class of a modulo m , which is $\bar{a} = a + m\mathbb{Z}$.

Theorem 1.2: Let $a, b, c, d \in \mathbb{Z}$ and $m \in \mathbb{N}$, with $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then we have the following statements are true:

1. $a + c \equiv b + d \pmod{m}$.
2. $a - c \equiv b - d \pmod{m}$.
3. $ac \equiv bd \pmod{m}$.

We denote $\mathbb{Z}/m\mathbb{Z}$, the set of all congruence classes modulo m .

2. Euler φ Function

Definition 2.3 (φ Function): The Euler φ function is defined as $\varphi : \mathbb{Z} \rightarrow \mathbb{N}$, where $\varphi(n) = \#\{x \in \llbracket 0, m-1 \rrbracket \mid \gcd(x, m) = 1\}$.

Lemma 4: Let $r, m, p \in \mathbb{N}$ where p is prime, then $\gcd(m, p^r) \neq 1 \Leftrightarrow p \mid m$

Proof. $\Rightarrow$ let $d = \gcd(m, p^r)$, since $d \mid p^r$ and p is prime then $d = p^k$ for some $k \in \mathbb{N}$, and $p^k = d \mid m$ thus $p \mid p^k$ and $p^k \mid m$ so $p \mid m$. $\Leftarrow$ suppose that $p \nmid m$ and since $p \nmid p^r$ too then $p \nmid \gcd(m, p^r)$ given $p > 1$ then $\gcd(m, p^r) \neq 1$. ■

Proposition 2.5: Let p be a prime number, then $\varphi(p) = p - 1$ and $\varphi(p^r) = p^r - p^{r-1} = p^r \left(1 - \frac{1}{p}\right)$.

Proof. Let $p, r \in \mathbb{N}$ with p prime. From the lemma, we have that for $m \in \mathbb{N}$, $\gcd(p^r, m) = 1 \Leftrightarrow p \nmid m$, the numbers that are divisible by p in $\llbracket 0, p^r - 1 \rrbracket$ are of the form kp where $k \in \llbracket 0, p^{r-1} - 1 \rrbracket$, so the number of numbers that are divisible by p in $\llbracket 0, p^r - 1 \rrbracket$ is $p^r - p^{r-1}$ so $\varphi(p^r) = p^r - p^{r-1}$. ■

Theorem 2.6 (Lagrange): Let G be a finite group and H a subgroup of G , then $\#H \mid \#G$.

This theorem is already done multiple times in algebra so no need to prove it here.

Lemma 2.7: Let $n \in \mathbb{N}$, $x \in (\mathbb{Z}/n\mathbb{Z})^*$ (invertible) $\Leftrightarrow \gcd(x, n) = 1$.

Proof. $\Rightarrow$ Suppose that $x \in (\mathbb{Z}/n\mathbb{Z})^*$, then $\exists y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = 1 \pmod{n}$, then $xy = 1 + kn \Rightarrow xy - kn = 1$ by Bezout theorem we have $\gcd(x, n) = 1$. $\Leftarrow$ Suppose that $\gcd(x, n) = 1$, by Bezout theorem, $\exists y, k \in \mathbb{Z}$, $xy + kn = 1$ and thus $xy + kn = 1 \pmod{n} \Rightarrow xy = 1 \pmod{n}$ so $x \in (\mathbb{Z}/n\mathbb{Z})^*$. ■

We need this lemma since $\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^*$.

Theorem 2.8 (Euler): Let m be a positive integer modulo a be an integer relatively prime to m then $a^{\varphi(m)} \equiv 1 \pmod{m}$.

Proof. Let a relatively prime to m , then $\gcd(a, m) = 1 \Rightarrow a \in (\mathbb{Z}/m\mathbb{Z})^*$, a has finite order thus $\exists k \in \mathbb{N}$, $a^k = 1 \pmod{m}$, $\langle a \rangle$ is a subgroup of $(\mathbb{Z}/m\mathbb{Z})^*$ so $k \mid \#(\mathbb{Z}/m\mathbb{Z})^* = \varphi(m)$ and thus $\varphi(m) = kb$, thus $a^{\varphi(m)} = a^{kb} = (a^k)^b = 1^b = 1 \pmod{m}$, which completes the proof. ■

Corollary 2.9 (Fermat): Let p be a prime, if the integer a is not divisible by p then $a^p \equiv a \pmod{p}$.

Proof. Just take $m = p$ in Euler's theorem, then $\varphi(m) = p - 1$ so $a^{\varphi(m)} = a^{p-1} \equiv 1 \pmod{p}$, so $a^p \equiv a \pmod{p}$. ■

3. Quadratic Residues

Definition 3.10 (Quadratic Residue): Let p be an odd prime and a an integer not divisible by p , we say that a is a quadratic residue modulo p if there exists $x \in \mathbb{Z}$, such that $x^2 \equiv a \pmod{p}$.

Theorem 3.11: An integer a is a quadratic residue modulo p if and only if $\gcd(a, p) = 1$ and a has a square rest modulo p .

Proof. $\Rightarrow$ Suppose that a is a quadratic residue modulo p , that is, there exists $x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{p}$. If $\gcd(a, p) \neq 1$ then $p \mid a$ so $a \equiv 0 \pmod{p}$ thus $x = 0$, so necessarily $\gcd(a, p) = 1$ and a has x^2 remainder when divided by p thus the rest is a perfect square. $\Leftarrow$ Suppose now that $\gcd(a, p) = 1$ and a has a square rest modulo p , then $a \not\equiv 0 \pmod{p}$ and $a \equiv x^2 \pmod{p}$ thus a is quadratic residue. ■

Definition 3.12 (Legendre Symbol): Let p be an odd prime, define the Legendre symbol as:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is not a quadratic residue} \\ 0 & \text{if } p \text{ divides } a \end{cases}$$

Theorem 3.13: Let p be an odd prime, for every integer a

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p}$$

Proof. We treat by cases

1. $p \mid a$: then $a \equiv 0 \pmod{p} \Rightarrow a^{\frac{p-1}{2}} \equiv 0^{\frac{p-1}{2}} \equiv 0 \pmod{p}$.
2. $p \nmid a$: then $\gcd(a, p) = 1$, by Fermat's little theorem we have that $a^{p-1} \equiv 1 \pmod{p}$, the square roots of 1 are -1 and 1 only thus $a^{\frac{p-1}{2}} \equiv 1$ or $-1 \pmod{p}$.
 1. If a is a quadratic residue then $\exists x \in \mathbb{Z}$, $x^2 \equiv a \pmod{p} \Rightarrow a^{\frac{p-1}{2}} \equiv x^{2 \cdot \frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \pmod{p}$.
 2. If $a^{\frac{p-1}{2}} \equiv -1 \pmod{p}$, we have that $\mathbb{Z}_p$ is a finite field thus there exists a generator g of $(\mathbb{Z}_p)^*$ (primitive element theorem in finite fields), then $a \equiv g^k \pmod{p}$, we have then $a^{\frac{p-1}{2}} \equiv g^{k \cdot \frac{p-1}{2}} \equiv -1 \pmod{p}$. Given that the generator g has order $p - 1$ then $k \cdot \frac{p-1}{2} \equiv 0 \pmod{p - 1}$ thus $k \cdot \frac{p-1}{2} = m(p - 1) \Rightarrow k = 2m$ so k is even, thus $a \equiv g^{2m} = (g^m)^2 \pmod{p}$ so a is a quadratic residue modulo p .

Given that a is a quadratic residue modulo p if and only if $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$ and $a^{\frac{p-1}{2}} \equiv -1$ or $-1 \pmod{p}$ then necessarily if a is not a quadratic residue, $a \equiv -1 \pmod{p}$. ■

Exercise 3.14:

1. Decompose into partial fractions in $\mathbb{R}[x]$ the rational function

$$\frac{x}{x^4 + x^2 + 1}$$

2. Let K be a commutative field, and let $p : x^2 + \lambda x + \mu$ be a monic polynomial of degree 2, show that p is reducible over K if and only if it has a root in K .
3. Let $K = \mathbb{Z}/5\mathbb{Z}$ be the field of residue classes, factor the polynomial $(x^2 + 4)(x^2 + 3)$ into irreducible factors over K .
4. Still with $K = \mathbb{Z}/5\mathbb{Z}$, decompose into partial fractions the rational function

$$\frac{x - \bar{2}}{(x^2 + \bar{4})(x^2 + \bar{3})}$$

1. Decomposing into partial fractions: we have that the denominator decomposes into the following two irreducible polynomials

$$\begin{aligned}
x^4 + x^2 + 1 &= x^4 + 2x^2 + 1 - x^2 \\
&= (x^2 + 1)^2 - x^2 \\
&= (x^2 + x + 1)(x^2 - x + 1)
\end{aligned}$$

and thus we have that the fraction decomposes as follows

$$\frac{x}{x^4 + x^2 + 1} = \frac{ax + b}{x^2 + x + 1} + \frac{cx + d}{x^2 - x + 1}$$

by multiplying both sides by $x^4 + x^2 + 1$ we get

$$\begin{aligned}
x &= (ax + b)(x^2 - x + 1) + (cx + d)(x^2 + x + 1) \\
0 &= ax^3 - ax^2 + ax + bx^2 - bx + b + cx^3 + cx^2 + cx + dx^2 + dx + d - x \\
0 &= (a + c)x^3 + (-a + b + c + d)x^2 + (a - b + c + d - 1)x + (b + d)
\end{aligned}$$

$$\Rightarrow \begin{cases} a + c = 0 \\ -a + b + c + d = 0 \\ a - b + c + d - 1 = 0 \\ b + d = 0 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = -1/2 \\ c = 0 \\ d = 1/2 \end{cases}$$

thus we get

$$\frac{x}{x^4 + x^2 + 1} = \frac{1}{2(x^2 - x + 1)} - \frac{1}{2(x^2 + x + 1)}$$

- Let K be a commutative field and let $P(x) = x^2 + \lambda x + \mu$. $\Rightarrow$ Suppose that $P(x)$ is reducible, then $P(x) = Q(x) \cdot R(x)$ with $\deg(Q) \neq 0$ and $\deg(R) \neq 0$ and since $\deg(Q) + \deg(R) = \deg(P)$ then necessarily $\deg(Q) = \deg(R) = 1$, so $Q(x) = ax + b$ and $R(x) = cx + d$ and since $a, b \in K$ then $-\frac{a}{b} \in K$ which is a root of Q and thus a root of P so P has a root in K . $\Leftarrow$ Suppose that $P(x)$ has a root α , then $(x - \alpha)$ divides P and thus we get that P is reducible.
- Let $K = \mathbb{Z}/5\mathbb{Z}$, factor $(x^2 + 4)(x^2 + 3)$ into irreducible factors over K , $-3 = 2 \bmod 5$ is not a quadratic residue since $2^{\frac{5-1}{2}} = 4 = -1 \bmod 5$ thus $x^2 + 3$ is irreducible given it has no roots. While $-4 = 1$ is a quadratic residue with roots -1 and 1 so $(x^2 + 4)(x^2 + 3) = (x^2 - 1)(x^2 + 3) = (x - 1)(x + 1)(x^2 + 3)$.
- Decomposing now the partial fraction in $\mathbb{Z}/5\mathbb{Z}$,

$$\begin{aligned}
\frac{x - 2}{(x^2 + 4)(x^2 + 3)} &= \frac{x - 2}{(x + 4)(x + 1)(x^2 + 3)} \\
&= \frac{a}{x + 4} + \frac{b}{x + 1} + \frac{cx + d}{x^2 + 3}
\end{aligned}$$

by multiplying all parts by $(x^2 + 4)(x^2 + 3)$ we get

$$x - 2 = a(x + 1)(x^2 + 3) + b(x + 4)(x^2 + 3) + (cx + d)(x + 4)(x + 1)$$

we set the following values for x

- $x = -1$: $-3 = b(-1 + 4)(1 + 3) = 12b = 2b \Rightarrow b = 3 * (-3) = -9 = 1$.
- $x = +1$: $-1 = a(1 + 1)(1 + 3) = 8a = 3a \Rightarrow a = 2 * (-1) = -2 = 3$.

Exercise 3.15:

- Decompose 561 into prime factors.
- Let a be an indeterminate, for any integer $n \geq 1$, justify the formula

$$a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \dots + a + 1)$$

- Justify for any integer $m \geq 1$ and any integer $n \geq 1$ the formula

$$a^{mn} - 1 = (a^m - 1)(a^{(n-1)m} + a^{(n-2)m} + \dots + a^m + 1)$$

- Let $a \in \mathbb{Z}$ be an integer, prove $a^{561} - a = a(a^{2 \cdot 280} - 1) = a(a^2 - 1)m$ for some integer m .
- For any $a \in \mathbb{Z}$, show that $a(a - 1)(a + 1)$ is divisible by 3.
- Show that $a^{561} - a$ is a multiple of 3.
- Show that $a^{561} - a$ is a multiple of 17.
- Show that $a^{561} - a$ is a multiple of 11.
- Show that $a^{561} \equiv a \bmod 561$.

- The prime decomposition of 561 is $3 \cdot 11 \cdot 17$.
- Let a be a formal variable, we have

$$(a - 1) \sum_{i=0}^{n-1} a^i = \sum_{i=0}^{n-1} a^{i+1} - a^i = \sum_{i=1}^n a^i - \sum_{i=0}^{n-1} a^i = a^n - 1$$

- By taking $a = a^m$ in the previous inequality we get the result.

4. Clearly $561 = 2 \cdot 280 + 1$ thus $a^{561} - a = a(a^{560} - 1) = a(a^{2 \cdot 280} - 1)$ and we have from the previous question by taking $m = 2$ and $n = 280$ that $a^{561} - a = a(a^2 - 1) \left(\sum_{i=1}^{279} a^{2i} \right) = a(a^2 - 1)m$.
5. Let $a \in \mathbb{Z}$, we have that $a = 3q + r$ with $0 \leq r < 3$ from the Euclidean division algorithm, then $a(a-1)(a+1) = (3q+r)(3q+(r-1))(3(q+1)+(r-2))$ and since $0 \leq r < 3$ then necessarily either r , $r-1$ or $r-2$ is 0, and thus it has one of the factors divisible by 3.
6. $a^{561} - a$ is divisible by 3 since $a^{561} - a = a(a+1)(a-1)m$ which is divisible by 3 from the previous question.
7. $a^{16} = 1 \pmod{17}$ by Fermat little theorem, then $a^{560} = a^{16 \cdot 35} = 1 \pmod{17}$ thus, $a^{561} = a \pmod{17} \Rightarrow 17 \mid a^{561} - a$.
8. Same here, $a^{10} = 1 \pmod{11}$ by Fermat little theorem, then $a^{560} = a^{10 \cdot 56} = 1 \pmod{11}$ thus, $a^{561} = a \pmod{11} \Rightarrow 11 \mid a^{561} - a$.
9. Given that all the divisors of 561 divide $a^{561} - a$ then it is divisible by their product so $561 \mid a^{561} - a \Rightarrow a^{561} \equiv a \pmod{561}$.

Chapter 1

Introduction To Information Theory

The digital revolution has transformed how we communicate, store and process information. This chapter introduces the fundamental concepts of information theory, a discipline at the intersection of mathematics, computer science and telecommunication.



This diagram presents the general structure of any communication system. The communication process begins with an information source. We start by thinking what a message could be. In general, we can consider a message to be constructed by a set of symbols $\mathcal{X} = \{x_1, \dots, x_n\}$ and given a context, has a distribution of which of those symbols are sent. Thus, we consider a message consecutive realizations of a random variable X defined in $\mathcal{X}$.

Now the goal is to quantify information, thus given $X \sim P_X(x)$, we define the amount of information for some symbol x to be $I(x) = -\log_2(P_X(x))$ (justifications for these definitions will be in the next chapter). Then, we can define an average amount of information from a source X , which is called the entropy $H(X) = \mathbb{E}[I(X)]$. Such a quantity is important given Shannon's first theorem, which gives a lower bound for any data compression.

After being able to measure the amount of information, we can measure efficiency of some encodings both for source and channel by how much redundancy is added to how much is actually needed.

1.1. The Communication System Model

Shannon's groundbreaking contribution was a formal model that describes communication between two entities, whether machine or humans. This model highlights the essential components:

- Source: the originator of the information.
- Channel: The medium through which information is transmitted.
- Recipient: The receiver of the information.

1.1.1. Source & Message

The sources generate a message composed of symbols from a specific alphabet.

- Discrete data: text, numbers, symbols.
- Digitized analog data: image, voice, video converted to digital form.

1.1.2. Transmitter Processing

The transmitter performs three crucial operations.

1. Source Coding: data compression to minimize message size.
2. Channel Coding: adding redundancy for error correction.
3. Modulation: converting digital messages to physical signal.

1.1.3. Receiver Processing

The receiver performs these receiver operations.

1. Demodulation: converting physical signal into a digital message.
2. Channel Decoding: error detection and correction.
3. Source Decoding: reconstruction of the original message.

Chapter 2

Information & Entropy

Information theory is a theory founded by Claude Shannon, providing a mathematical framework for quantifying information, studying its transmission, processing, and storage. This chapter focuses on the fundamental concepts of information measurement through entropy and related measures.

2.1. Quantifying Information

Notation: Let Σ be a set:

- $\Sigma^n = \Sigma \times \dots \times \Sigma$ is the set of words of length n . We take that $\Sigma^0 = \{\varepsilon\}$ where ε is the empty word.
- $\Sigma^* = \bigcup_{i \geq 0} \Sigma^i$ the set of all possible words over Σ .
- $\Sigma^+ = \bigcup_{i \geq 1} \Sigma^i$ the set of all possible non-empty words over Σ .
- For $x \in \Sigma^*$, we define $|x| = \min\{n \in \mathbb{N} | x \in \Sigma^n\}$.
- The set of binary letters as $\mathbb{B} = \{0, 1\}$.
- For $x = x_1 \dots x_n, y = y_1 \dots y_m \in \Sigma^*, xy = x_1 \dots x_n y_1 \dots y_m \in \Sigma^*$.

Definition 1 (Symbol/Message): Let $\mathcal{X} = \{x_1, \dots, x_m\}$ be a finite set of alphabets. An element of $\mathcal{X}$ is called a symbol, and elements of $\mathcal{X}^+$ are called a message with alphabet $\mathcal{X}$.

To model information that was given from a message, we consider the fact that a message contains more information the more unpredictable it is. For example, consider a sender that keeps sending the message “Hello!” over and over again, at each iteration, it becomes predictable what the next message would be, thus there is no more information that is passed. Uncertainty is a property of a random process, thus a good way to model a message is to consider it as a random variable X that has values in $\mathcal{X}$. From the previous example too, we

can think that the information in a message is related to how uncertain it is, that is, how less probable it would be sent. Also, notice that the information from independent events should be additive, that is, if even p and q happen that are independent, then the amount of information gained from the even pq happening is the sum of information gained from both the events p and q .

Notation: For the sake of simplifying notation, we will denote $P(X = x)$ as $P(x)$, same for other random variables, like $P(Y = y)$ as $P(y)$. Same for all the remaining notations like $P(y|x)$ which is $P(Y = y | X = x)$.

To define a measure for information $I : \mathcal{X} \rightarrow \mathbb{R}$, we start by considering some random variable X that represents a source of symbols $\mathcal{X}$. As discussed before, the information will be a function of probability, thus we take $I(x) = \varphi(P(x))$ from our previous discussion it would satisfy the following postulates:

- Postulate 1: Information depends only on probability $I(A) = \varphi(P(A))$.
- Postulate 2: Certain events provide no information $\varphi(1) = 0$.
- Postulate 3: Information decreases with increasing probability, φ is decreasing.
- Postulate 4: Impossible events provide infinite information $P(0) = \infty$.
- Postulate 5: Information from independent events $\varphi(pq) = \varphi(p) + \varphi(q)$.

Also, clearly φ would be continuous since a tiny difference in probability would result in a tiny difference in the amount of information that it would result in.

We will briefly prove that $\varphi \equiv -\log_b$ for any base b . By induction we have that $\varphi(a^n) = n\varphi(a)$ and since $\varphi(a) = \varphi(a^{q/q}) = q\varphi(a^q)$ then necessarily $\varphi(a^{p/q}) = (p/q)\varphi(a)$. By continuity, we have $\forall x \in \mathbb{R}, \varphi(a^x) = x\varphi(a)$. Now take $t = \log_a x$, then $\varphi(x) = \varphi(a^t) = t\varphi(a) = \varphi(a) \log_a x$, thus $\varphi(x) = \varphi(a) \log_a(x)$, by taking $x \rightarrow 0$, we have that $\varphi(a) \log_a(x) \rightarrow +\infty$ so $\varphi(a) < 0$. Therefore, $\varphi \equiv -\log_b$ for some b .

Definition 2 (Measure Of Information): Let X be a random variable that represents the probability that a message x is sent as $P(X = x) = P(x)$, we define the measure of information of the message x as

$$I_D(x) = -\log_D P(x)$$

2.2. Measures Of Entropy

After we defined a measure of information, which we discussed to be a measure of uncertainty, we need a way to quantify the average amount of uncertainty for all the values of X . Thus, we define the entropy as follows.

Definition 3 (Entropy): Let X be a random variable with values in $\mathcal{X}$, we define the entropy $H(X)$ as the average amount of information, that is

$$H_D(X) = \mathbb{E}[I_D(X)] = - \sum_{x \in \mathcal{X}} P(x) \cdot \log_D P(x)$$

If D is not specified then we consider $D = 2$ in general. Notice that the information of some message x depends only on its unpredictability, that is, its probability of occurrence $P(x)$ not its value.

Proposition 4: Let X, Y with symbols in $\mathcal{X}, \mathcal{Y}$ be two sources.

1. Non-negativity: $H_D(X) \geq 0$.
2. Maximum Entropy: $H_D(X) \leq \log_D \# \mathcal{X}$.
3. Independent Additivity: $X \perp Y \Rightarrow H_D(XY) = H_D(X) + H_D(Y)$.

Proof.

1. $\forall x \in \mathcal{X}, I(x) \geq 0$, then $H(X) = \mathbb{E}[I(X)] \geq 0$
2. Here, we optimize on the distribution of X , $\max_P H(X) = -\min_P \sum_{x \in \mathcal{X}} P(x) \cdot \log_2 P(x)$, hence for I a finite set $\min_{\{y_i \in [0,1]\}} \sum_{i \in I} y_i \log_2(y_i)$ with $\sum_{i \in I} y_i = 1$. Notice that all the terms have the same sign so $\min \sum_{i \in I} \dots = \sum_{i \in I} \min \dots$ y_i all having the same range would mean that we just need to find c such that $y_i = c$ and satisfies the second inequality, which is clearly $c = \frac{1}{\#I}$. So X would be a uniform distribution over $\mathcal{X}$, $H_u(X) = \mathbb{E}[I(X)] = -\sum_{x \in \mathcal{X}} 1/(\# \mathcal{X}) \log_2(1/(\# \mathcal{X})) = \log_2 \# \mathcal{X} \geq H(X)$.
3. Let X, Y be two independent variables, then $H(XY) = \mathbb{E}[I(XY)] = \mathbb{E}[I(X) + I(Y)] = \mathbb{E}[I(X)] + \mathbb{E}[I(Y)] = H(X) + H(Y)$. ■

Example: Let $\mathcal{X} = \{0, 1\}$, and X a discrete random variable with values in $\mathcal{X}$, which has the distribution

$$P(x) = \begin{cases} 0.9 & \text{if } x = 0 \\ 0.1 & \text{if } x = 1 \end{cases} \Rightarrow I(x) = \begin{cases} 0.152 & \text{if } x = 0 \\ 3.321 & \text{if } x = 1 \end{cases}$$

And the measure of average amount of information is $H(X) = 0.4689$.

Example: DNA sequences provide an excellent example of entropy application.

Let's consider:

- Primate DNA: $P(A) = P(G) = P(C) = P(T) = 1/4$. $H(X) = 2$
- Bacterial DNA: $P(G) = P(C) = 0.38$, $P(A) = P(T) = 0.12$. $H(X) = 1.795$.

The entropy of bacterial DNA indicates more predictable sequences compared to the primate DNA.

2.2.1. Joint Entropy

Definition 5 (Joint Entropy): Let X, Y be two random variables with joint distribution $P(x, y)$, the joint entropy is

$$H_D(X, Y) = - \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P(x, y) \log_D P(x, y) = -\mathbb{E}[\log_D P(X, Y)]$$

Proposition 6: Let X, Y be two sources.

1. Symmetry: $H_D(X, Y) = H_D(Y, X)$.
2. Non-Negativity: $H_D(X, Y) \geq 0$.
3. Upperbound: $H_D(X, Y) \leq H_D(X) + H_D(Y)$.
4. Lowerbound: $H_D(X, Y) \geq \max(H_D(X), H_D(Y))$.

Proof.

1. trivial by $P(x, y) = P(X = x, Y = y) = P(Y = y, X = x) = P(y, x)$.
2. $\forall x \in \mathcal{X}, \forall y \in \mathcal{Y}, 0 \leq P(x, y) \leq 1 \Rightarrow \log_D P(x, y) \leq 0$ so we have $P(x, y) \log_D P(x, y) \leq 0$. Hence,

$$H_D(X, Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P(x, y) \log_D P(x, y) \geq 0$$

3. We have $H_D(X, Y) - H_D(X) - H_D(Y) = \sum_{x \in \mathcal{X}} P(x) \log_D P(x) + \sum_{y \in \mathcal{Y}} P(y) \log_D P(y) - \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} P(x, y) \log_D P(x, y)$, by using the

fact that $P(x) = \sum_{y \in Y} P(x, y)$, we can transform all the sums to be indexed with $X \times Y$ as follows:

$$\begin{aligned} \sum_{x \in X} P(x) \log_D P(x) &= \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x) \\ \sum_{y \in Y} P(y) \log_D P(y) &= \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(y) \end{aligned}$$

Thus we obtain the writing

$$H_D(X, Y) - H_D(X) - H_D(Y) = \sum_{(x, y) \in X \times Y} P(x, y) \log_D \left(\frac{P(x)P(y)}{P(x, y)} \right)$$

We have that $\forall t \geq 0, \log_D(t) \leq t - 1$ thus

$$\begin{aligned} H_D(X, Y) - H_D(X) - H_D(Y) &= \sum_{(x, y) \in X \times Y} P(x, y) \log_D \left(\frac{P(x)P(y)}{P(x, y)} \right) \\ &\leq \sum_{(x, y) \in X \times Y} P(x, y) \left(\frac{P(x)P(y)}{P(x, y)} - 1 \right) \\ &\leq \sum_{(x, y) \in X \times Y} P(x)P(y) - P(x, y) \\ &\leq \sum_{(x, y) \in X \times Y} P(x)P(y) - \sum_{(x, y) \in X \times Y} P(x, y) \\ &\leq 1 - 1 = 0 \end{aligned}$$

Therefore

$$H_D(X, Y) \leq H_D(X) + H_D(Y)$$

4. Without loss of generality, we prove that $H_D(X, Y) \geq H_D(X)$, we have from before that $P(x) = \sum_{y \in Y} P(x, y)$ and since $\forall (x, y) \in X \times Y, P(x, y) \geq 0$ then $\forall (x, y) \in X \times Y, P(x, y) \leq P(x)$.

$$\begin{aligned} H_D(X, Y) &= - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x, y) \\ &\geq - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x) = H_D(X) \end{aligned}$$

and same to prove that $H_D(X, Y) \geq H_D(Y)$ and hence we obtain the inequality $H_D(X, Y) \geq \max(H_D(X), H_D(Y))$. ■

2.2.2. Conditional Entropy

The other useful measure of entropy is the conditional entropy, which measures the amount of information remaining in a random variable knowing another.

Definition 7 (Conditional Entropy): Let X, Y be two random variables with joint distribution $P(x, y)$, the conditional entropy is

$$H_D(X|Y) = - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x|y) = -\mathbb{E}[\log_D P(X|Y)]$$

Proposition 8: Let X, Y be two sources.

1. *Non-Negativity:* $H_D(X|Y) \geq 0$.
2. *Chain Rule:* $H_D(X, Y) = H_D(X) + H_D(Y|X)$.
3. *Reduction Of Uncertainty:* $H_D(X|Y) \leq H_D(X)$.

Proof.

1. $\forall (x, y) \in X \times Y, 0 < P(x|y) \leq 1 \Rightarrow \log_D P(x|y) \leq 0 \Rightarrow H_D(X|Y) = - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x|y) \geq 0$.
2. We have

$$\begin{aligned} H_D(X, Y) &= - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x, y) \\ &= - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(x) - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(y|x) \\ &= - \sum_{x \in X} \left(\sum_{y \in Y} P(x, y) \right) \log_D P(x) - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(y|x) \\ &= - \sum_{x \in X} P(x) \log_D P(x) - \sum_{(x, y) \in X \times Y} P(x, y) \log_D P(y|x) \\ &= H_D(X) + H_D(Y|X) \end{aligned}$$

3. $H_D(X|Y) \stackrel{a}{=} H_D(X, Y) - H_D(Y) \stackrel{b}{\leq} H_D(X) + H_D(Y) - H_D(Y) = H_D(X)$, where a comes from the chain rule and b from the subadditivity of joint entropy. ■

2.2.3. Mutual Information

Conditional entropy measures the amount of information is remaining in a random variable knowing another, it is easy to see that by subtracting the amount of information a variable has with the amount of remaining information knowing another we get the information that is “shared” between them.

Definition 9 (Mutual Information): Let X, Y be two sources, we define the mutual information as

$$I_D(X; Y) = \mathbb{E} \left[\log_D \left(\frac{P(X, Y)}{P(X)P(Y)} \right) \right] = \sum_{(x, y)} P(x, y) \log_D \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

Proposition 10: Let X, Y be two sources.

1. *Equivalent Formulas:*

$$\begin{aligned} I_D(X; Y) &= H_D(X) - H_D(X|Y) \\ &= H_D(Y) - H_D(Y|X) \\ &= H_D(X) + H_D(Y) - H_D(X, Y) \\ &= H_D(X, Y) - H_D(X|Y) - H_D(Y|X) \end{aligned}$$

2. *Symmetry:* $I_D(X; Y) = I_D(Y; X)$.

3. *Non-Negativity:* $I_D(X; Y) \geq 0$.

4. *Upperbound:* $I_D(X; Y) \leq \min(H_D(X), H_D(Y))$.

5. *Self-Information:* $I_D(X; X) = H_D(X)$.

Proof.

1. Trivial, just write the formulas.
2. Direct consequence of the first two formulas of 1.
3. We have from the reduction of uncertainty that $H_D(X|Y) \leq H_D(X)$ thus $I_D(X; Y) = H_D(X) - H_D(X|Y) \geq 0$.

4. We have from the lowerbound of joint entropy that $H_D(X, Y) \geq \max(H_D(X), H_D(Y))$ then

$$\begin{aligned} I_D(X; Y) &= H_D(X) + H_D(Y) - H_D(X, Y) \\ &\leq H_D(X) + H_D(Y) - \max(H_D(X), H_D(Y)) = \min(H_D(X), H_D(Y)). \end{aligned}$$

5. $I_D(X; X) = H_D(X) - H_D(X|X) = H_D(X)$. ■

2.2.4. Data-Processing Inequality

Another important property is the data processing inequality, stating informally that, given some information, one cannot process it to obtain more information. So, we start with defining what is “data-processing”.

Definition 11 (Markov Chain): Let X, Y, Z be three random variables, we say that they form a Markov chain in that order, and we denote it $X \rightarrow Y \rightarrow Z$ if the conditional probability of Z depends only on Y and not on X , which can be written as

$$P(z|x, y) = P(z|y)$$

This definition is natural, since in our case, we take that we have the two random variables X and Y , and we produce Z with some processing on Y without knowing anything about X , the process can be deterministic or not.

To prove the data-processing inequality, we need the following lemma.

Lemma: $I_D(X; Y, Z) = I_D(X; Y) + I_D(X; Z|Y)$ where

- $I_D(X; Y, Z) = \mathbb{E}[\log_D(P(X, Y, Z)/(P(X)P(Y, Z)))]$.
- $I_D(X; Z|Y) = \mathbb{E}[\log_D(P(X, Z|Y)/(P(X|Y)P(Z|Y)))]$.

It is left for the reader to prove, it is mainly algebraic manipulation.

Theorem 12 (Data-Processing Inequality): Let X, Y, Z be three sources such that $X \rightarrow Y \rightarrow Z$, we have that

$$I_D(X; Y) \geq I_D(X; Z).$$

Proof. Since $X \rightarrow Y \rightarrow Z$ then there is no extra information given from Z knowing Y thus $I_D(X; Z|Y) = 0$ we have

$$I_D(X; Y, Z) = I_D(X; Y) + I_D(X; Z|Y)$$

$$I_D(X; Y, Z) = I_D(X; Z) + I_D(X; Y|Z)$$

hence

$$I_D(X; Y) + I_D(X; Z|Y) = I_D(X; Z) + I_D(X; Y|Z)$$

$$\Rightarrow I_D(X; Y) = I_D(X; Z) + I_D(X; Y|Z)$$

since $I_D(X; Y|Z) \geq 0$ then $I_D(X; Y) \geq I_D(X; Z)$. ■

Chapter 3

Source Coding

Source coding is a fundamental part of communication that deals with representing data to be sent in the most compressible manner to get an efficient communication. From Shannon's theorem, we will see that the amount of data compression is bounded, and thus we will see algorithms that achieve an average length 1 bit away from the bound.

3.1. Source Coding

The most important part of coding theory is achieving the most efficient reliable and secure coding. The first part is the focus of this section, by assigning a code from each symbol in $\mathcal{X}$, we try to achieve the minimum bound possible of letters to send to transmit our message.

Definition 1 (Coding Function): Let $c : \mathcal{X} \rightarrow \mathbb{D}^+$, we call it a coding function, which takes characters of our set of symbols, and represent it as a string in $\mathbb{D}^+$ which is the set of words with D alphabets.

To measure the efficiency of our coding function, we define the average code length, the less the average, the more efficient the transmission will be.

Definition 2 (Average Code Length): Let $c : \mathcal{X} \rightarrow \mathbb{D}^+$, a coding function, and consider the function

$$\bar{L} : \{c : \mathcal{X} \rightarrow \mathbb{D}^+\} \rightarrow \mathbb{R} \quad c \mapsto \bar{L}_c = \sum_{x \in \mathcal{X}} p(x) \cdot |c(x)|$$

Example: Take a horse race with 8 horses, we want to send a message in binary that indicates which horse has won. And suppose that the probabilities of winning for each horse is as follows

Horse	1	2	3	4	5	6	7	8
Probability Of Winning	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$

If we just send the index of the winner horse, we get the following coding function

Horse i	1	2	3	4	5	6	7	8
$c_1(i)$	000	001	010	011	100	101	110	111

If we do it just blindly, we will need 3 bits to describe all the possible winner horses from 000, 001, 010, ..., 110, 111, calculating the average of description length we get

$$\bar{L}_{c_1} = \sum_{x \in X} 3 \cdot p(x) = 3 \text{ bits/sym}$$

Giving us an average of 3 bits for the transmission to give exactly who horse is the winner. An aspect that we did not use in the previous part is how likely do horses win in this case, which we can use to improve the amount of bits that would be sent on the channel.

Notice that if we assign a smaller message for the horses that are more probable to win, and longer message to horses that are less likely to win, then we can reduce some of the data that will be used to specify the winner. We take the following encoding

Horse i	1	2	3	4	5	6	7	8
$c_2(i)$	0	10	110	1110	111100	111101	1111110	111111

Now, we recalculate the average description length to get

$$\bar{L}_{c_2} = \sum_{x \in X} |c(x)| \cdot p(x) = 2 \text{ bits/sym}$$

we have reduced the average description length by 1 bit. If we calculate the entropy in this case we get $H(X) = -\sum_{x \in X} P(x) \cdot \log_2 P(x) = 1.83475$, we see that the reduced average code length is more than the entropy.

Definition 3 (Properties Of Codes): Let $c : \mathcal{X} \rightarrow \mathbb{D}^+$ be a coding function. We say that it is:

- *Non-Singular:* if $\forall x, y \in \mathcal{X}, c(x) \neq c(y)$.
- *Uniquely Decodable:* if $\forall y_1, \dots, y_n \in \mathbb{D}^+, \exists! x_1, \dots, x_m, y_1 y_2 \dots y_n = c(x_1) c(x_2) \dots c(x_m)$, that is, any sequence $y_1, \dots, y_n$ of $\mathbb{D}^+$ can be uniquely decoded into a sequence $x_1, \dots, x_m$ of $\mathcal{X}$.
- *Instantaneous/Prefix-Free:* if $\forall x, y \in \mathcal{X}, \nexists m \in \mathbb{D}^+, x = ym$, that is, no code-word is a prefix of another code-word.

3.1.1. Principles Of Instantaneous Code Construction

Lemma 4 (Kraft Inequality): Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be a set of symbols, and c be a coding function, set $l_i = |c(x_i)|$ the length of the coding of the i^{th} symbol, then the following statements are true.

1. If c is an instantaneous code then $\sum_{i=1}^n D^{-l_i} \leq 1$.
2. If c satisfies $\sum_{i=1}^n D^{-l_i} \leq 1$, then there exists an instantaneous code $c' : \mathcal{X} \rightarrow \mathbb{D}^+$ such that $\forall i \in [1, n], |c'(x_i)| = |c(x_i)|$.

Theorem 5 (Shannon's First Theorem): Let X be a random variable with values in $\mathcal{X}$, for any coding function $c : \mathcal{X} \rightarrow \mathbb{D}^+$ we have

$$\bar{L}_c \geq H_D(X)$$

and there exists a prefix code $c' : \mathcal{X} \rightarrow \mathbb{D}^+$ such that

$$H_D(X) \leq \bar{L}_{c'} \leq H_D(X) + 1$$

Algorithm 5.1 (Shannon-Fanon):

1. Sort symbols in decreasing order of probability.
2. Partition the set into two subsets with approximately equal probability.

3. Assign 0 to one subset and 1 to the other.
4. Recursively apply steps 2-3 to each subset.

Huffman's algorithm provides a systematic way to construct an optimal prefix code, where symbols with higher probability should have shorter code words, and achieves a lower bound of average coding weight.

Algorithm 5.2 (Huffman):

1. Sort symbols by probability in descending order.
2. While there is more than one node:
 1. Find two nodes with smallest probability.
 2. Create a new node with these children, the probability of the node is the sum of the probabilities of the children.
 3. Assign 0 to one branch and 1 to another.
3. Recursively apply the steps 1-2 until only one node remains.

Chapter 4

Channel Coding

Coding has two parts, the source coding and channel coding, the first handles how to compress data to get the least redundancy possible while containing all information to get exactly the same message. Channel coding on the other hand, adds redundancy in way to make it more resilient to noise affecting the signal.

4.1. What Is A Channel?

Suppose you have a transmission medium, where given a stream of symbols X^n from an alphabet $\mathcal{X}$, it can transport it to a stream of symbols Y^n . The way it transmits can be affected by randomness. We will call the transmission medium a channel, as it takes information from an input and moves it to an output.

To formalize the idea of a channel, we can think of a simple example. Suppose you are typing on a keypad, you will represent the sender, the keypad would be the receiver, and your hand would be the channel. There is a non-zero chance that you will do a mistake and write an unwanted symbol, what we would like to know, is the probability of for example clicking 3 by mistake when you wanted to type 6. If we represent the wanted key as a random variable X and the pressed key as Y , the question becomes really easy in this case, we just want to calculate the probability $P(Y = 3|X = 6)$. So, in general, when considering any channel, it is best to know what is the behavior of the conditional probability $P(Y|X)$.

Definition 4.1.1 (Channel): A channel is a conditional probability $P(y|x)$, describing the probability of receiving an output y when sending an input x called the transition probability.

Definition 4.1.2 (Discrete Channel): A channel is said to be discrete if the random variables describing the input and output both are discrete, that

is, they have values in finite sets $\mathcal{X} = \{x_1, \dots, x_n\}$ and $\mathcal{Y} = \{y_1, \dots, y_m\}$ of symbols, in this case, the transition probability is a matrix of the following form

$$Q = \begin{pmatrix} P(y_1|x_1) & P(y_2|x_1) & \dots & P(y_m|x_1) \\ P(y_1|x_2) & P(y_2|x_2) & \dots & P(y_m|x_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(y_1|x_n) & P(y_2|x_n) & \dots & P(y_m|x_n) \end{pmatrix}$$

and we define the channel as the triplet $\mathcal{C} = (\mathcal{X}, P(Y|X), \mathcal{Y})$.

Definition 4.1.3 (Discrete Memoryless Channel): A discrete channel is said to be memoryless if for $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ we have

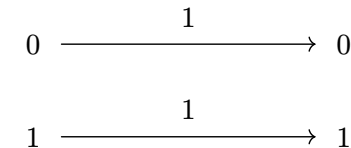
$$P(y|x) = \prod_{i=1}^n P(y_i|x_i)$$

that is, the distribution of the output only depends on the input and not previous channel inputs or outputs.

Most of the remaining of the course will focus mainly on discrete memoryless channels as in they are the simplest types of channels and are more enough for practical purposes in general.

Example:

- **Binary Noiseless Channel:** a noiseless channel is a channel where we have that $P(y = b|x = b) = 1$, that is, the transmission is perfect.

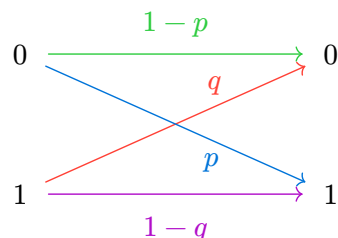


- **Binary Useless Channel:** given any input, it returns 0, it is useless as it can't pass information, which will reexplained later.

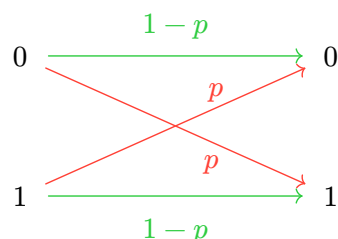
- **Noisy Binary Channel:** given $0 \leq p, q < 1$ and we have the following transition matrix

$$Q = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}$$

this channel represents a binary channel where 0 has probability $1 - p$ of getting flipped to one, and 1 has probability $1 - q$ of getting flipped to zero.

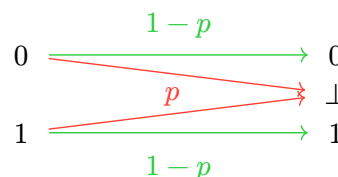


- **Binary Symmetric Channel:** this channel represents the most intuitive idea of a noisy binary channel, where we consider just a noisy binary channel with $p = q$, that is, the probability of getting a flip for 1 and 0 is the same.



- **Binary Erasure Channel:** this channel models the phenomenon of having an erasure or a corruption of data in the channel, where we have a way to recognize where the erasure has happened. We consider $\mathcal{X} = \{0, 1\}$ and $\mathcal{Y} = \{0, 1, \perp\}$ where $\perp$ denotes the erasure, and the following transition matrix.

$$Q = \begin{pmatrix} 1-p & p & 0 \\ 0 & p & 1-p \end{pmatrix}$$



4.2. Noise & Redundancy

We have seen some examples of channels, the question that comes to mind is how and when is a reliable communication establishable. Consider the binary symmetric channel with probability p of flipping a bit. When sending a single bit x , there is a chance that the received bit would be $\bar{x}$. To avoid such an error, we can think of sending the same bit multiple times and then do a majority vote to decide which bit has been sent. So the encoding function is $x \mapsto x^n$ for some fixed n , and the decoding function is $(y_1, \dots, y_n) \mapsto 1$ if $\#1(y_i) > \#0(y_i)$ and 0 otherwise. Notice now that when we have 1 error and $n \geq 3$, we can determine and correct the error just with those encoding and decoding functions. From Algebra & Coding lectures, we know that we can detect up to $\frac{n}{2}$ errors, and thus, the probability of actually making an error in a BSC is

$$P_e(n) = \sum_{k=\lceil n/2 \rceil}^n \binom{n}{k} p^k (1-p)^{n-k}.$$

By taking $p = 0.1$ for example we get the following probabilities for n errors.

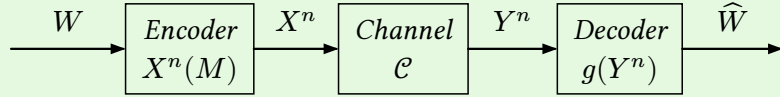
n	1	3	5	7	9
$P_e(n)$	0.1	0.0280	0.0085	0.0027	0.0008

All beautiful, P_e is decreasing by Chernoff's bound and converges to 0 when n tends to infinity. But notice that we have a trade-off, to reduce the possibility for an error happening, we reduced in the rate of transmission, that is, to just receive one symbol, we need to send n bits, thus giving us a rate of $1/n$ bits per symbol. The problem with this scheme is that the more errors we want to fix, we lose rate. The question now is the following "If we take a rate R , how much can we reduce the probability of errors.", which is answered by Shannon's Channel Coding Theorem.

Before that, we formalize the ideas and the question we have until now with the following definitions. We start by defining a communication system using those channels, which will represent how a real communication system would look and interact like and how codes work in such a system.

Definition 4.2.4 (Communication System): Let $\mathcal{C} = (\mathcal{X}, P(Y|X), \mathcal{Y})$ be a discrete channel.

- A message W drawn from an index set $\llbracket 1, M \rrbracket$.
- A sender sequence $X^n(W)$.
- A receiver sequence $Y^n \sim P(Y^n|X^n)$.
- A decoding function $\hat{W} = g(Y^n)$.



An error happens when $\hat{W} \neq W$.

Definition 4.2.5 ((M, n)-Code): Let $\mathcal{C} = (\mathcal{X}, P(Y|X), \mathcal{Y})$ be a discrete channel, an (M, n) -code consists of the following:

- An index set $\llbracket 1, M \rrbracket$.
- An encoding function $X^n : \llbracket 1, M \rrbracket \rightarrow \mathcal{X}^n$, which gives codewords $x^n(1), x^n(2), \dots, x^n(M)$, we call the set of codewords the codebook.
- A decoding function $g : \mathcal{Y}^n \rightarrow \llbracket 1, M \rrbracket$ which assigns a guess $\hat{W}$ of what message was sent W .

We define the rate of the code $R = \log(M)/n$.

The rate of an (M, n) code represents how many bits of information are needed for a single message M to be sent.

Example:

- Consider for example a set of messages with $M = 2$ and $n = 10$, what we mean is that for each message we need a block of 10 symbols in the transmission to send 1 bit, if we calculate the rate it would be $\frac{\log(2)}{10} = \frac{1}{10}$ bit per transmission, so we take 9 extra bits to counter the noise from the channel.
- We take the previously stated example of repetition code, with $n = 3$, we have that $M = 2$ since we just send a bit, which has values 0 or 1, the encoding function would be as follows $x^3(0) = 000$ and $x^3(1) = 111$ and g to be a majority vote decoder, the rate would be $\frac{\log(2)}{3} = \frac{1}{3}$ bit per transmission which is indeed what we found before.

Definition 4.2.6 (Conditional Probability Error): Consider an (M, n) -code, we define the conditional probability error of index $i \in \llbracket 1, M \rrbracket$ as

$$\lambda_i^{(n)} = P(g(Y^n) \neq i | X^n = x^n(i))$$

The conditional probability is meant to represent the probability of getting an error on message i , which naturally gives us a way to define what is the most probable error we can get while decoding.

Definition 4.2.7 (Maximal/Average Probability Of Error): The maximal probability of error $\lambda^{(n)}$ for an (M, n) -code is

$$\lambda^{(n)} = \max_{i \in \llbracket 1, M \rrbracket} \lambda_i^{(n)}$$

and the average probability of error $P_e(n)$ as

$$P_e(n) = \frac{1}{M} \sum_{i=1}^M \lambda_i^{(n)}$$

Definition 4.2.8 (Achievable Rate): A rate R is said to be achievable if there exists a sequence of $(2^{nR}, n)$ -codes such that $\lambda^{(n)} \xrightarrow{n \rightarrow \infty} 0$.

Definition 4.2.9 (Channel Rate Capacity): The rate capacity of a channel is the supremum of all achievable rates.

Now we have all definitions needed, we can describe the problem in a formal way.

Which rates R are achievable?

4.3. Channel Information Capacity

As we discussed before, the mutual information of two variables X and Y represents how much information is shared between the two variables X and Y . Statistically speaking, this represents how much Y correlates with X . If the channel is noiseless, we have that $I(X; Y) = H(X)$ which signifies that all the

information has been passed without any noise, while $I(X; Y) = 0$ just means that Y is independent of X thus no information has been passed at all.

Notice that the more information that can be passed, the better the channel is. But given a channel, we can never change $P(y|x)$ since it comes from the physical implementation of the channel. So the only parameter we have to improve the mutual information in a channel is the distribution of X , since $I(X; Y) = H(X) - H(X|Y)$. Thus, we define the following bound.

Definition 4.3.10 (Channel Information Capacity): Let $\mathcal{C}$ be a channel, X, Y represent the random variables of input and output respectively. We define the channel capacity as the distribution of X that maximizes the information that can be passed through it, that is,

$$C_{\mathcal{C}} = \max_{P(x)} I(X; Y)$$

To appreciate this definition, we consider the following two examples. Start with the useless binary channel $\mathcal{C}$, we have that $P(Y = 0|X = x) = 1$ for any x and thus $I(X; Y) = 0$ since $X \perp Y$, therefore $C_{\mathcal{C}} = 0$, hence this channel is useless, since it indeed cannot send any information. Now suppose we have a noiseless channel $\mathcal{C}'$, and a sender always sends $X = 0$, we will also have $I(X; Y) = 0$, which is a misuse of the channel. To get the best from the channel, we consider a distribution that maximizes the information.

Example:

- **Capacity Of Binary Symmetric Channel:** Consider a BSC $\mathcal{C}$ with probability of a bit flip p , we have

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum_{x \in X} P(x) H(Y|X = x) \\ &\stackrel{(a)}{=} H(Y) - \sum_{x \in X} P(x) H(p) \\ &= H(Y) - H(p) \stackrel{(b)}{\leq} 1 - H(p) \end{aligned}$$

where in (a) we used the fact that $Y|X = x$ is a Bernoulli distribution and thus its entropy is $H(p) = -p \log p - (1-p) \log(1-p)$ and (b) the fact that Y is a binary random variable thus $H(Y) \leq 1$ which is attained only if Y has a uniform distribution. And thus we get that $C_{\mathcal{C}} = 1 - H(p)$.

- **Capacity Of Binary Erasure Channel:** Consider a BEC $\mathcal{C}$ with probability p of having an erasure, we have

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum_{x \in X} P(x) H(Y|X = x) \\ &= H(Y) - \sum_{x \in X} P(x) H(p) = H(Y) - H(p) \end{aligned}$$

If we denote $\pi = P(X = 1)$, we have that $H(Y) = H(p) + (1-p)H(\pi)$, and hence $I(X; Y) = (1-p)H(\pi) \leq 1-p$ with equality satisfied when $\pi = \frac{1}{2}$, so the capacity is $C_{\mathcal{C}} = 1-p$.

Proposition 4.3.11: Let $\mathcal{C}(X \rightarrow Y)$ be a channel and C the capacity of $\mathcal{C}$.

- $0 \leq C \leq \min(\log \# X, \log \# Y)$.
- $I(X; Y)$ is continuous and concave function of $P(x)$.
- The capacity is well defined since $I(X; Y)$ is upper bounded for $P(x)$ and a local maximum is global by the concavity.

4.4. Typical Sequences & AEP

To discuss the theorem, we have to define a type of sequences that come from the Asymptotic Equipartition Property, which are the typical sequences.

Definition 4.4.12 (Typical Sequence): Let $X_1, \dots, X_n \sim P(x)$ be an i.i.d random variables with entropy $H(X)$, and let $\varepsilon > 0$ fixed.

$$A_{\varepsilon}^{(n)} = \left\{ x^n \in \mathcal{X}^n \mid \left| \frac{1}{n} \log P(x^n) - H(X) \right| < \varepsilon \right\}.$$

A ε -typical sequence is an element $x^n \in A_{\varepsilon}^{(n)}$.

Theorem 4.4.13 (Asymptotic Equipartition Property): Let a sequence of i.i.d random variables $X_1, \dots, X_n \sim P(x)$ with entropy $H(X)$, $\forall \varepsilon > 0$:

- If $x_n \in A_\varepsilon^{(n)}$ then $2^{-n(H(X)+\varepsilon)} \leq P(x^n) \leq 2^{-n(H(X)-\varepsilon)}$.
- $P(A_\varepsilon^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$.
- $(1 - \delta)2^{n(H(X)-\varepsilon)} \leq \# A_\varepsilon^{(n)} \leq 2^{n(H(X)+\varepsilon)}$ for large n and any $\delta > 0$.

Example: When the sources generates a long sequence of symbols, the Law Of Large Numbers guarantees that it will almost surely be a typical sequence. A more intuitive way to think of it is to consider a coin toss, with $P(1) = 0.9, P(0) = 0.1$, and consider the two following sequences:

$$s = 11100111010000110010101100101011101010001000111001$$

$$s' = 11111111111011110001110111111111111111111111111111$$

notice that s' is more likely to be generated by the distribution P than s , this is what the idea a typical sequence tries to capture, sequence that are in essence resemble the distribution. If we calculate the number of typical sequences we get that the number of typical sequences for $n = 100$, we get approximately $2^{100H(0.9)} = 2^{100 \cdot 0.46890...} \simeq 2^{46.89}$, while the number of actual sequences is 2^{100} , notice that $\frac{2^{46.89}}{2^{100}} = 2^{-53.1}$ thus we have that a really small set of sequences actually might be the outcome, while the others are extremely unlikely, asymptotically.

4.5. Channel Coding Theorem

From the Asymptotic Equipartition Property, we have that for each typical n -sequence generated with X , there are approximately $2^{nH(Y|X)}$ possible Y sequences which are all equiprobable with probability around $2^{-nH(Y|X)}$, now if we consider the set of typical n -sequences generated with Y , we have that there are around $2^{nH(Y)}$ typical sequences. In our encoding, we would like the set of sequences of Y generated from the noise of the channel after sending a sequence in X to be disjoint so we have no ambiguity in decoding the sequence to the original message, the total number of such disjoint sets is $\frac{2^{nH(Y)}}{2^{nH(Y|X)}} = 2^{nI(X;Y)}$, and hence we can send at most around $2^{nI(X;Y)}$ distinct sequences. To maximize $2^{nI(X;Y)}$ is to maximize $I(X;Y)$ which we know is exactly the information capacity of the channel. But also, notice that $2^{nI(X;Y)}$ is really close to the form 2^{nR} , indeed, if

an (M, n) -code is being sent through the channel, then $M \leq 2^{nI(X;Y)}$ to be able to decode without confusion, and thus

$$R = \frac{\log(M)}{n} \leq \frac{\log(2^{nI(X;Y)})}{n} = I(X;Y)$$

, thus necessarily the rate would be at most the information sent through the channel. Conversely, if we take $R > C$ then $M = 2^{nR} > 2^{nC}$ codewords, and from our discussion before, we know that we can have at most 2^{nC} disjoint sets, so in this case we will indeed get two sets that are not disjoint and hence, not achieve the rate with errors going to 0 since the element in the intersection of the codewords sent from two different x vectors would always be confused.

Theorem 4.5.14 (Channel Coding Theorem): Let $\mathcal{C}$ be a discrete memoryless channel.

- **Achievability:** any rate $R < C_{\mathcal{C}}$ where $C_{\mathcal{C}}$ is the information capacity of the channel $\mathcal{C}$ is achievable, that is, there exists a sequence of $(2^{nR}, n)$ -codes with maximum probability error $\lambda^{(n)} \xrightarrow{n \rightarrow \infty} 0$.
- **Converse:** any sequence of $(2^{nR}, n)$ -codes satisfying $\lambda^{(n)} \xrightarrow{n \rightarrow \infty} 0$ must satisfy $R \leq C$.